

Manuel Capiendo

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EDUCATION

Florida Atlantic University — Bachelor of Science

Expected December 2026

Major: **Computer Science** | Minor: **Artificial Intelligence** | GPA: 3.32

Relevant Coursework: Data Structures & Algorithms, Software Engineering, Database Structures, Artificial Intelligence, Machine Learning

Certifications: AWS Certified Cloud Practitioner, Lean Six Sigma Yellow Belt

SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, C

Data & Search: PostgreSQL, MongoDB, OpenSearch

Frameworks & Libraries: React.js, Angular, Flask, Node.js

Tools: Postman, Git, GitHub, Jira, Agile

EXPERIENCE

NextEra Energy | Software Development Engineer Intern

May 2026 – August 2026

- Engineered a pipeline that extracted asset and condition detections from AI-model outputs, translated them into business-facing labels, and synced that data from PostgreSQL into OpenSearch to power search and filtering on NextVision, a company-wide imagery platform.
- Built the corresponding filtering and search UI in React and TypeScript, including advanced filter menus and shareable deep links, with unit and end-to-end tests to validate functionality.

Universal Destinations & Experiences | Software Quality Engineering Intern

January 2026 – April 2026

- Extended and maintained mobile automation frameworks to validate new app updates across Orlando, California, and Japan properties, using Java, Python, Selenium, and Appium to catch regressions before release.
- Developed and executed test plans for web and eCommerce platforms, including cross-browser and cross-device validation using LambdaTest across multiple release cycles and device types.

NextEra Energy | Programmer Analyst Intern

September 2025 – December 2025

- Joined a new initiative to centralize NextEra's 18,000+ internal nuclear APIs and built an AI-powered search feature using Angular, Flask, Python, and OpenAI's LLMs to identify relevant APIs from natural language requests; estimated to reduce discovery time by roughly 60%.
- Led a refactor of a difficult-to-maintain front-end architecture, then presented the finished platform to the director and nuclear development team, who adopted it as the foundation for NextEra's nuclear IT development standards.

NextEra Energy | Nuclear Solutions Analyst Intern

May 2025 – August 2025

- Built autonomous patrol routes for robotic quadrupeds at the St. Lucie Nuclear Power Plant, developing timed inspection pathways for automated facility monitoring; personally presented the project to regional Nuclear Regulatory Commission directors.
- Researched and built a VR proof-of-concept control room prototype using Unity and Meta Quest with LiDAR-based 3D environments, exploring virtual operator training as an alternative to on-site visits.

Sparton Corporation | Acoustic Engineering Intern

May 2024 – August 2024

- Identified a testing bottleneck in sonobuoy production causing delays and led the initiative to solve it, coordinating with multiple departments and customizing existing impedance analyzer driver code in C to eliminate an advanced operator to run manual tests.
- Integrated the automated test station with a SQL Server database to store test results, reducing testing cycle time for piezoelectric ceramic components from 160 to 40 seconds (75% improvement) and restoring full-capacity, on-schedule line operation.

PROJECTS

NEXTSteps: Career Navigation App – NEXT100 Hackathon

July 2025

- Led a 6-person team of IT interns over a week-long hackathon to design an internal career navigation tool that uses the company's organizational chart to identify employees with similar skills and backgrounds; voted Fan Favorite.

AI-Powered Waste Routing – Hackathon (2nd Place, Solo)

August 2024

- Proposed modeling garbage truck routing and waste-to-energy facilities after how ants forage for food, applying Ant Colony Optimization to real-world waste collection logistics. Competed solo against team-based competitors, placing 2nd overall.

MENTORSHIP

Amazon Fulfillment Technologies & Robotics | Amazon Guides Mentee

September 2025 – January 2026

- Selected for a structured mentorship program pairing students with Amazon engineering leadership for weekly technical sessions on fulfillment center robotics architecture, workcell automation, and large-scale systems design.

Coding Instructor Volunteer | Flamingo Coders, South Florida Tech Hub

February 2025 – May 2025

- Volunteered monthly alongside another instructor to teach K-12 students fundamental programming concepts using Code.org and freeCodeCamp, guiding students through problem-solving tailored to their skill level.